Muhtasim Haque Nahian

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Website | LinkedIn | ResearchGate | Google Scholar | ORCID

Research Interests

Atmospheric science; hydrodynamic and hydrologic modeling; nature-based solutions; remote sensing.

Education

Louisiana State University Aug 2026 – Present
M.S. in Oceanography and Coastal Sciences
Louisiana, United States; first semester in progress.

Bangladesh University of Engineering and Technology
B.Sc. in Water Resources Engineering Apr 2019 – July 2024
Dhaka, Bangladesh; CGPA: 3.32/4.00.

Work Experience

Graduate Research Assistant Aug 2026 – Present
Department of Oceanography and Coastal Sciences, Louisiana State University, Louisiana, United States.

Research Assistant July 2024 – June 2026
Department of Water Resources Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.

Intern Nov 2023
Water Supply, Sanitation and Urban Water Management Division, Institute of Water Modelling, Dhaka, Bangladesh.

Thesis Research

Master's Thesis — Ongoing
Advancing Seasonal Weather Prediction Using CESM2 Large Ensemble Climate Simulations.
Supervisor: Dr. Wengui Liang, Assistant Professor, DOCS, LSU.

Undergraduate Thesis
Rain-on-Grid Modeling of Bagerhat Paurashava using HEC-RAS 2D.
Supervisor: Dr. Md. Mostafa Ali, Professor, DWRE, BUET.

Conference Proceedings

1. Nahian, M. H., & Rahman, M. A. (2025). Prediction of Above-Ground Biomass Using Machine Learning Algorithm and Comparative Analysis Across Different Time Periods. *10th International Conference on Water and Flood Management (ICWFM 2025)*, IWFM, Dhaka, Bangladesh.

2. Sarker, P. K., Bhoumik, P. K., & **Nahian, M. H.** (2025). Dynamic Analysis and Forecasting of the Effect of Changing Land Use on Ecosystem Service Value: A CA-ANN Model Based Approach. *10th International Conference on Water and Flood Management (ICWFM 2025)*, IWFM, Dhaka, Bangladesh.

3. Nahian, M. H., Akhand, N. U. H., Sagar, S. D., Tabassum, R., & Arafat, M. (2024). Development of a Hydrologic Model for Flood Prediction and Suitability Analysis of Boro Crop Cultivation for Sunamganj Haor Region Using HEC-HMS. *7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024)*, Khulna, Bangladesh. (Under review in *Journal of Engineering Science*.)

4. Rahman, M. A., & **Nahian, M. H.** (2023). Integration of AHP Method in Flood Risk Mapping of Pabna Using GIS and Multicriteria Analysis. *9th International Conference on Water and Flood Management (ICWFM 2023)*, IWFM, Dhaka, Bangladesh.

5. Nahian, M. H., Sara, S., Turjo, M., Rahman, M. A., & Mashuk, F. (2023). A Case Study of Rahmatkhali Riverbank Erosion: Comparison Among Riparian Management, Coir Logs and Geo Bags. *4th International Conference on Planning, Architecture & Civil Engineering (ICPACE 2023)*, RUET, Rajshahi, Bangladesh.

Poster Presentations

1. Nahian, M. H., & Zahid, S. M. T. (2025). Application of the SCS-CN Method for Runoff Estimation using Google Earth Engine in the GBM Basin of Bangladesh. *AGU25*.

2. Turjo, M., & **Nahian, M. H.** (2025). Future Hydro-Meteorological Responses in the Barak River Basin under CMIP6 Ensemble Scenarios. *AGU25*.

3. Zahid, S. M. T., Moulik, H., Sarker, M. M., Khan, M. F., & **Nahian, M. H.** (2025). Flood Susceptibility Assessment in the Chittagong Hill Tracts (CHT), Bangladesh, Using GIS-Based AHP and Multicriteria Analysis. *AGU25*.

Projects

River Halda Hydro-Biological Modeling Feb – June 2026
Development of Hydro-Biological Models for River Basin Management in River Halda for Better Management of Natural Spawning of Fishes.

Economic Corridor Development Study Sep 2025 – Jan 2026
Economic Corridor Development Study of Bangladesh, funded by ADB.

Coastal Towns Drainage Planning July 2024 – Aug 2025
Integrated Drainage Plan and Detail Design (IDPDD) Services for Five Coastal Towns under Coastal Towns Climate Resilience Project (CTCRP), LGED, funded by ADB.

Academic Projects

Hydrologic Modeling of Sunamganj Haor Sep 2023
Development of a Hydrologic Model for Flood Prediction and Suitability Mapping of Boro Crop Cultivation for Sunamganj Haor Region Using HEC-HMS.
Course: WRE 404 — Mathematical Modeling in Water Resources Engineering-II.

Comparison of 1D and 2D Hydrodynamic Models Feb 2023
Comparison of 1D vs 2D Hydrodynamic Model: Contraction of Width using Venturi Flume.
Course: WRE 304 — Mathematical Modeling in Water Resources Engineering-I.

Technical Skills

Water modeling: HEC-RAS, HEC-HMS, PC-SWMM, EPA SWMM, SWAT.

Habitat modeling: PHABSIM, River2D.

Geospatial analysis: ArcGIS, QGIS, Google Earth Engine, Google Earth.

Programming: Python, MATLAB.

Application tools: Office365, G-suite, LaTeX.

Honors & Awards

Champion, Water Challenge 2.0, BUET WRE FEST 2023.

Gold Award, International Poster Competition on Achieving SDG in Developing Countries 2023 by Univ.

2nd Runner-up, Hult Prize On-Campus Round, Hult Prize Foundation.

Finalist, Mindshift 1.3: Set the Mark (2019), Engineering Students' Association of Bangladesh.

Board Scholarship in all school certificate examinations.

Award of Merit, St. Joseph Higher Secondary School, Dhaka, Bangladesh, for securing first place in examinations.
Award of Merit, Ahmed Bawany Academy, Dhaka, Bangladesh, for securing first place in examinations.

Leadership

President, Kantho:BUET, 2023–2024

Vice-President, BUET Brainiacs, 2023–2024

Organizer, BUET WRE FEST 2023, Department of Water Resources Engineering, BUET.

Editor, Yearly magazine published by the Department of Water Resources Engineering, BUET.

Test Scores

IELTS: 7.5 overall
Listening: 7.5; Reading: 7.5; Writing: 7.0; Speaking: 7.0.

GRE: 312 total
Quantitative: 161; Verbal: 151; Analytical Writing: 3.5.

References

Dr. Wengui Liang
Assistant Professor, DOCS, Louisiana State University, Louisiana, United States.
Academic Advisor and Thesis Supervisor.
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Dr. Md. Mostafa Ali
Professor, DWRE, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.
Undergraduate Thesis Supervisor and Project Supervisor.
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